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Late Carboniferous–Late Permian Paleomagnetic overprinting of Carboniferous granitoids in southern Bohemian Massif (Austria)

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Abstract

Paleomagnetic investigations on Early–Middle Carboniferous granitoids of the Austrian part of the South Bohemian Pluton (48.5°N, 14.2°E) yielded two groups of characteristic remanent magnetization (ChRM) directions, one being characterized by SSW-declinations and positive inclinations and labelled A1 ($D = 200°$, $I = 14°$, $k = 67$, $N = 9$; virtual geomagnetic pole (VGP) 31°N, 170°E), the other group having the same declinations but with negative inclinations ($D = 201°$, $I = -21°$, $k = 88$, $N = 18$; VGP 48°N, 162°E). Both magnetizations are considered as magnetic overprints owing to distinct recrystallization processes that occurred in different hydrothermal and tectonic conditions in Late Carboniferous and Late Permian times. In some cases the original magnetizations may remain after the remagnetization processes, but as a whole this study confirms that the paleomagnetic history of a magmatic crystalline basement (in this case of Late Carboniferous to Late Permian age) maybe differ significantly from its intrusion history (in this case Early to Middle or Late Carboniferous).

1. Introduction

The western half of the Austrian section of the Bohemian Massif (Fig. 1) is occupied by the granitoids of a Late Paleozoic magmatic complex referred to as the South Bohemian Pluton (SBP). The variety of intrusives exposed in the area can be divided into an older (Early Carboniferous), synorogenic and a younger (Late Carboniferous), late- to postorogenic group of granitoids (Finger et al., 1988; Frasl and Finger, 1991). A recent comprehensive investigation of the whole magmatic sequence accessible in the Austrian part of the SBP was intended to complete the paleomagnetic data available for the Upper Paleozoic in the Bohemian Massif (e.g. Krs, 1968; Soffel and Harzer, 1991) in order to follow the geotectonic evolution of the massif, as well as to contribute to the knowledge of Late Variscan overprinting phenomena in Central European (e.g. Edel, 1987).

The disadvantage of using intrusives for paleomagnetic investigations is the lack of tectonic control. On the other hand, the thermal history can be followed using different radiometric dating methods, so that attempts can be made to date magnetizations. Previous investigations carried out in different Permo-Carboniferous basins

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of the Bohemian Massif (Birkenmaier et al., 1968; Krs, 1968) have encountered flat-lying units which exclude post-Early Permian tilting. Preliminary paleomagnetic results (Reisinger et al., 1993) demonstrate that most of the magnetizations are Late Carboniferous–Permian overprints. The aim of this paper is to present these overprints and to discuss their implications.

2. Geology and sampling

The most frequent rock types in the SBP are granites and granodiorites, locally associated with small dioritic to quartzmonzodioritic bodies (Finger et al., 1988). Geochemical features classifying the granitoids into S-, I- and I/S-transitional-types are indicative of the Variscan plate

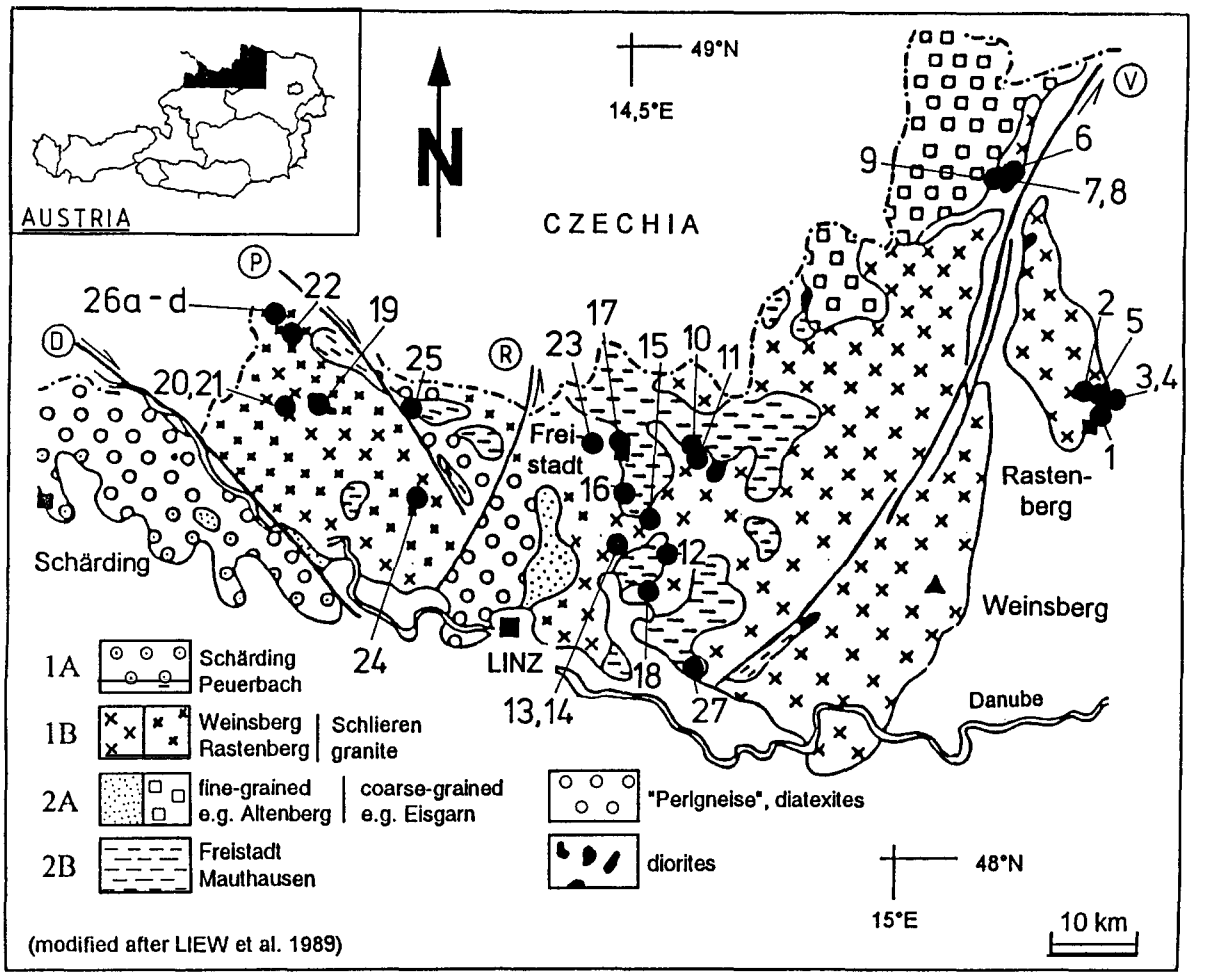


Fig. 1. Geological sketch map of the Austrian part of the South Bohemian Pluton (modified after Liew et al. (1989) with dots indicating the locations of paleomagnetic sampling sites (1–27). Legend column with combined numbers and letters refers to older (1) resp. younger (2) generations of granitoids, letters A/B denote plutonic subdivisions with S-type/I-type characteristics. Major dextral resp. sinistral strike–slip faults are Danube (D) and Pfahl (P) resp. Rodl (R) and Vitis (V) shear zones. Insert map shows the position of the study area within Austrian territory.

tectonic environment (Liew et al., 1989). In this classification I-type means an igneous type, broadly corresponding to the biotite–hornblende tonalite association and S-type means sedimentary type, broadly corresponding to the two mica granite association. Interpretation of paleomagnetism requires a well-defined timetable of magmatic activity. The age relations between a few larger plutons and numerous small intrusive bodies are well known (Frasl and Finger, 1988, cum lit.). The oldest intrusives are divided into two magmatic types, the 1A, S-type granitoids which were not sampled here, and the 1B, I-type granitoids (Weinberg and Rastenberg granites) which

Table 1
Final results of the CRM for the BM

Site	Rock type	Group	max. tb	Comp.	Dec	Inc.	n	k	α_{95}
BM1	Granitoids and dykes cutting Dobragneiss	1B	250–400	A	200.6	−20.4	13	24.0	8.6
			250–400	A1	190.2	12.7	5	31.7	13.8
2	Rastenberg inclusion	1B	350–400	A	197.4	−11.5	7	76.6	6.9
3	Lamprophyr cutting Dobragneiss		350; 580	A	196.8	−20.5	7	16.6	15.3
			230–400	A	209.3	−1.5	4	80.5	10.3
4	Amphibolite layers in Dobragneiss		350–400	A	198.0	−17.8	4	26.1	18.3
			350–400	A1	190.7	17.8	8	23.8	11.6
5	Amphibolite		280–400	A1	210.8	26.2	4	14.7	23.9
7	Diorite	2A	350–400	A	197.5	−21.4	3	81.6	13.7
8	Granitic dyke		280–300	A1	198.3	23.7	2		
10	Freistadt granodiorite roof-anatect. paragn.	2B	250–400	A	197.8	−28.4	4	25.3	18.6
			250–400	A1	214.3	37.5	3	156.5	9.8
12	Diorite	1B	350–400	A	221.2	−32.3	2		
			550–580	A1	189.1	17.0	2		
13	Titane mottled diorite	2A	350–400	A	198.5	−34.3	24	55.0	4.0
			550–580	A	204.1	−13.5	9	26.0	10.3
14	Altenberg granite	2A	350; 580	A	205.8	−30.0	13	34.4	7.2
15	Migma-granite	2A	280–300	A	192.1	−12.2	2		
16	Freistadt granodiorite	2B	350; 580	A	208.9	−22.6	14	28.1	7.6
			350; 580	A1	196.5	11.5	2		
17	Freistadt granodiorite	2B	350; 600	A	204.3	−21.3	18	25.0	7.4
18	Diorite	1B?	350–580	A	203.9	−25.3	8	76.5	6.4
			350–400	A1	205.3	16.4	2		
19	Diorite	1B	550–580	A1	192.1	5.5	11	61.3	5.9
21	Weinsberg granite (qmd-variety)	1B	500–580	A	200.2	−11.8	5	51.4	10.8
			350; 580	A1	204.7	19.9	4	21.4	20.3
22	Quartzmonzodiorite (qmd), inclusions	2B	350–400	A	201.2	−25.8	22	54.1	4.4
			550–600	A1	205.8	15.1	5	50.8	10.8
26	Quartzmonzodiorite (qmd)	2B	350–400	A	201.9	−17.0	14	46.0	5.9
			550–580	A1	204.2	11.8	37	127.0	2.1
26ab	Qmd contact zones to dykes 26c, 26d	2B	350–400	A	197.3	−25.7	6	180.7	5.0
			550–580	A1	202.1	10.2	9	169.1	4.0
26c	Granitic dyke	2B	550–580	A1	206.7	9.8	5	72.6	9.0
26d	Aplitic dyke	2B	350–400	A	198.8	−26.1	5	221.1	5.2
27	Mauthausen granite	2B	350; 580	A	216.2	−8.7	2		
			280; 580	A1	208.2	29.4	4	33.7	16.1
Mean A direction				A	201.3	−20.9	18	87.9	3.7
Mean A1 direction				A1	200.4	14.7	9	66.7	6.3
UGP A				A	48.4N	162.1E	18	194.3	2.5
UGP A1				A1	31.0N	170.6E	9	94.6	5.2

Group is consistent with Fig. 1; max.tb means maximum unblocking temperature; Dec. and Inc. are the directions of the ChRM: n is the number of specimens; and k and α_{95} are the usual statistic parameters.

are considered to be syn-tectonic magmatic features. The youngest intrusives are also divided into two groups, successively the 2A, S-type granite (Altenberg granite), followed by the posttectonic, calcalcaline, I-type, 2B plutonic units. Radiometric dating by different methods established the absolute chronology of intrusions, and of later thermal and tectonic events.

For the older granitoids of the Rastenberg/Weinsberg unit (1B in Fig. 1.) U/Pb dating of monazite and zircon yielded Visean intrusion ages of respectively, 328 Myear and 337 Myear (Friedl et al., 1992a; Klötzli, 1993) consistent with the 331 Myear found on the Mrakotin granite in the Czech Republic which also belongs to the SBP (Matte et al., 1990). The U/Pb ages of 314 Myear, 316 Myear and 318, Myear, respectively, obtained from xenotime, zircon and monazite (Von Quadt and Finger, 1991) probably date to a post intrusion recrystallization. An upper Carboniferous cooling age for the 2B group is supported by a K/Ar dating on muscovite at 309 Myear for the Schlägl granite (Frank and Schabert, unpublished work) and an U/Pb age on monazite at 302 Myear for the Freistadt granite (Friedl et al., 1992b). Several other K/Ar ages from 2A and 2B granitoids (Frank and Schabert, unpublished work) cluster within the 298–279 Myear time span corresponding to the Carboniferous–Permian boundary. The consistency of these later dates with the Ar/Ar plateau ages of 281, 294 and 287 Myear obtained for mylonites from the Rodl and Danube strike–slip faults question whether they correspond to magmatic cooling ages or rather to a reset in relation to the Late Variscan tectonic activity. Summing up, the intrusion of the first generation (group 2A) is well dated and occurred in the range 340–328 Myear. The ages of 309 and 302 Myear obtained for the 2B group may represent cooling ages or a late event as is the case for the U/Pb dates on the Weinsberg granite.

Sampling was performed on the different granitoids and mainly on associated diorites (Table 1). The oldest diorites which are associated with the Weinsberg and Rastenberg granites were collected at sites BM 2 and BM 19. Another fine-grained diorite (BM13) with a mottled structure resulting from titanite phenocrysts, displaying discordant contact with the Weinsberg granite, is intruded by the Altenberg granite (BM14) in an outcrop described by Frasl and Finger (1988). The youngest diorites consist of small quartzmonzodioritic bodies (sites BM 22, 26) associated with Mauthausen-type granites intruded south of the Pfahl shear zone. In site BM 26 a more complex situation was open to sampling; an irregular network of fine- to medium-grained granitic dykes (BM 26c) with diffuse boundaries to the country rock penetrates the QMD intrusive body. Both are cut by aplitic (to aplogranitic) dykes (BM 26d) with sharp contacts.

3. Rock magnetism

3.1. Techniques

Depending on size of exposure a paleomagnetic site covers some meters squared to several tens of meters squared. If different granitoids were sampled in an outcrop (road cutting or quarry), each rock type was treated as an individual site. Sampling was done by drilling oriented cores in the field where the orientation was taken by a magnetic compass. The cored samples, 25.4 mm in diameter, were cut into 22 mm long specimens. Remanent magnetization measurements were performed on modified Digico equipment (Strasbourg) and on a 2G-cryogenic system (Leoben/Gams); in both laboratories a Digico susceptibility bridge was used for susceptibility determination. Thermal demagnetization was performed using laboratory-made ovens as well as commercial devices.

3.2. Magnetic properties

In granitic rocks the NRM intensities vary from 0.1 to 10 mA m^{-1}, with susceptibility values between 10^{-5} to 10^{-4} SI units. Rocks dioritic in composition are stronger magnetically with NRM intensities ranging from 1 to 100 mA m^{-1} and susceptibilities varying from 10^{-4} to 10^{-3} SI units. In pilot studies applying (AF) alternating field as well as thermal cleaning, the latter procedure

turned out to be more efficient and therefore was used as the standard method for stepwise demagnetization. Possible mineral changes resulting from heating were controlled by measuring susceptibility after each thermal step.

For several sites of diorites (BM 6, 7, 8, 23, 24) and one lamprophyre (BM 20), despite rather high intensities of NRM, the demagnetizing process could not isolate characteristic magnetizations. The magnetization decreases up to 200–

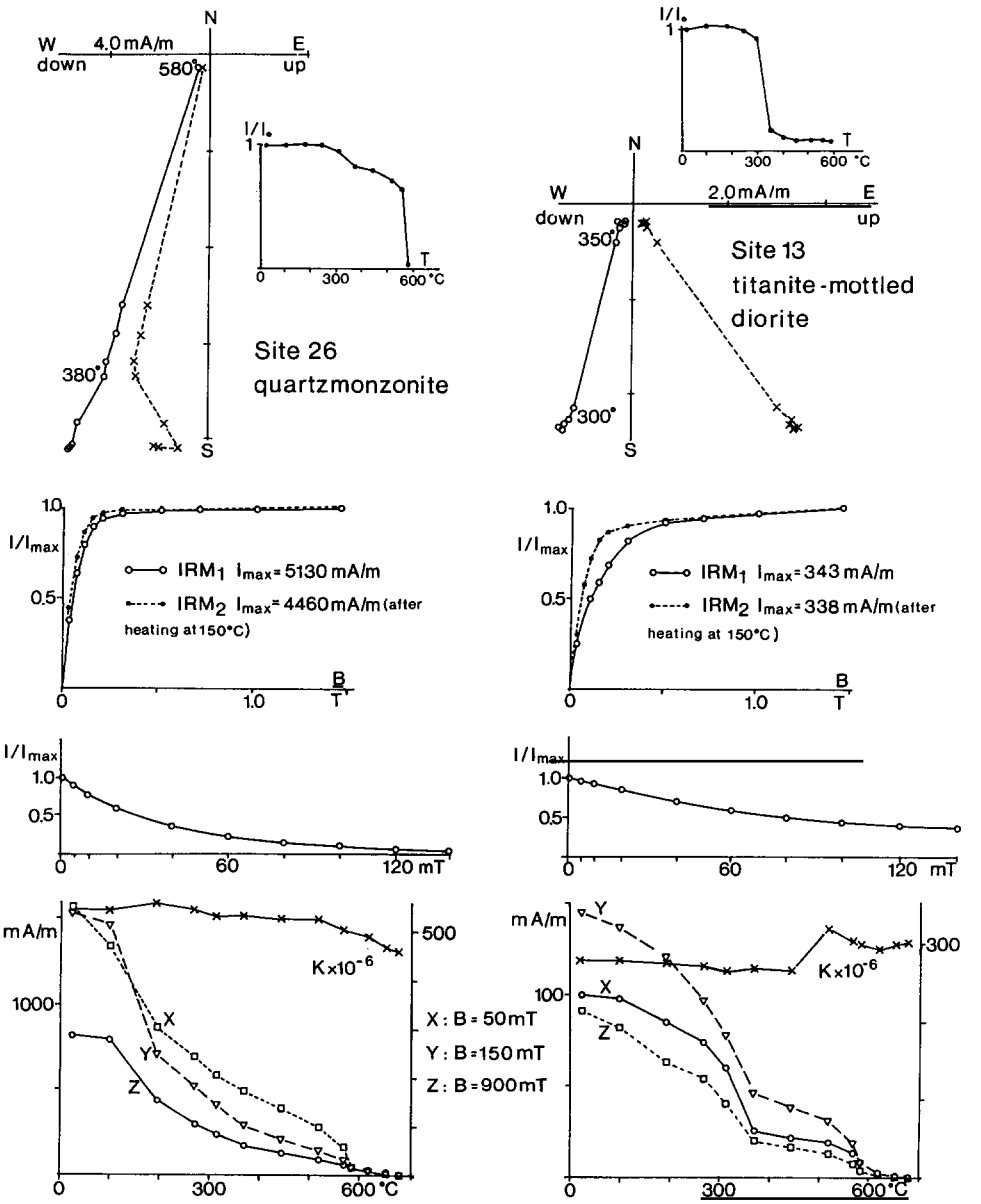


Fig. 2. Typical magnetic behavior in a quartzmonzonite (BM 26) and a titanite mottled diorite. (a) Thermal demagnetizing curves, Zijderveld projections (dots, horizontal projection; crosses, vertical north–south projection) and I/IO curve: (b) IRM curves before and after heating at 150°; (c) AF demagnetization of IRM; (d) thermal demagnetization of a three-component IRM, produced by magnetizing the core at 900 mT along its z-axis, then 150 mT along the y-axis and finally 50 mT along the x-axis (Lowrie, 1990).

280°C, then becomes unstable and increases drastically. The directions are random so that these sites had to be rejected. But most of the other rocks investigated display a two-component magnetization with corresponding unblocking temperatures in the range of (1) 350–400°C, and (2) 550–580°C (Fig. 2a, Table 1.). In rare cases a weak residual magnetization with unblocking temperatures > 600° was also observed.

The low-temperature component prevails in granitoids and diorites. Whereas the intensity of remanence shows a more or less dramatic decrease at the lower unblocking temperature, the magnetic susceptibility remains almost stable up to 600°C (BM 26, Fig. 2a) or increases slightly (BM, 13, Fig. 2d). This excludes (titano)maghemite as the carrier of the low-temperature component, the disintegration of which should be

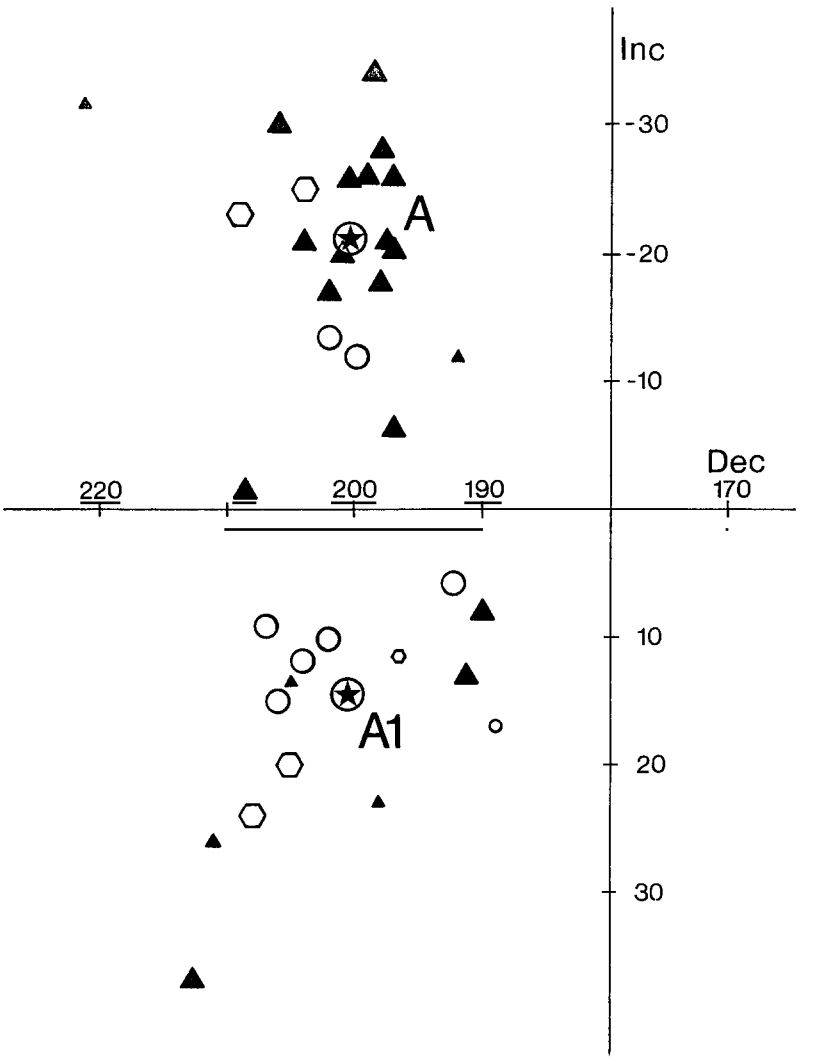


Fig. 3. Site mean directions listed in Table 1 plotted on an orthogonal net. Triangles represent low-temperature magnetizations (300–350°C); open dots represent components carried by magnetite; hexagons come from sites with both, low and intermediate unblocking temperatures. Mean directions for the A and A1 clusters are illustrated by stars.

indicated by a substantial drop in susceptibility (Soffel, 1991). On the other hand IRM-acquisition behavior (Fig. 2b) reveals the existence of a high-coercivity phase besides (titano)magnetite, even after heating to 150°C. In site 13 where a strong low-temperature component coexists with a weak magnetite component. AF treatment with a peak field up to 140 mT results in a demagnetization of only 50% of the initial IRM intensity (Fig. 2c). Thermal demagnetization of a three-component IRM (Lowrie, 1990) from this site shows the same behavior of the demagnetizing curves and a maximum of the intensity for the intermediate field of 150 mT (Fig. 2d). The unblocking temperatures around 330°C may indicate that pyrrhotite is responsible for the low-temperature magnetization (Dekkers, 1990). The slight increase of the susceptibility above 400°C may be due to oxidization of pyrrhotite to magnetite, but we do not observe the alteration of magnetite into hematite at higher temperatures. Microscopic observations on polished sections from site 13 have provided evidence of the presence of drop-like sulfides in titanites (M. Honnorez, personal communication, 1993) which may carry our low-temperature component.

The components with unblocking temperatures around 580°C are mostly residual in granitoids as well as in diorites. The opposite is the case in the quartzmonzodiorite BM 26 (Fig. 2). The low coercitivity and the unblocking temperature of 580°C of the dominating component indicate magnetite as the carrier of the magnetization. This is also confirmed by microscopic observations on polished sections (M. Honnorez, personal communication).

4. Paleomagnetic results

4.1. The directions

Owing to the lack of sediments in the granitoid terrain of the SBP only the in situ directions of remanence can be used for interpretation. The rocks investigated mostly exhibit two distinct directions of characteristic remanent magnetization (ChRM), one with SSW-declination and negative inclination labeled A, the other with SSW-declination and positive inclination labeled A1 (Table 1 and Fig. 2, BM 26). Both directions are usually characterized by different unblocking temperature spectra apart from a few exceptions; the low-temperature mineral, interpreted as pyrrhotite, carries the A direction while magnetite is mostly characteristic of the A1 direction (Fig. 3). Very rarely, both directions are identified in similar carriers, or the same direction can be carried by both sulfides and magnetite (Fig. 2, BM 13).

In various rock types, thermal demagnetization has indicated a few but stable magnetizations with directions significantly different from A and A1. When plotting the well-defined individual directions from different sites on the same stereonet, two main clusters with, respectively, northwesterly and southeasterly declinations and flat inclinations appear (Fig. 4). Furthermore, two less-substantiated clusters nearly opposite with east–northeasterly and west–southwesterly declinations and shallow inclinations also appear. These directions are labeled C (NW/SE) and B (ENE/WSW) referring to similar directions obtained in other Paleozoic massifs from Western Europe (Edel, 1987; Edel and Wickert, 1991).

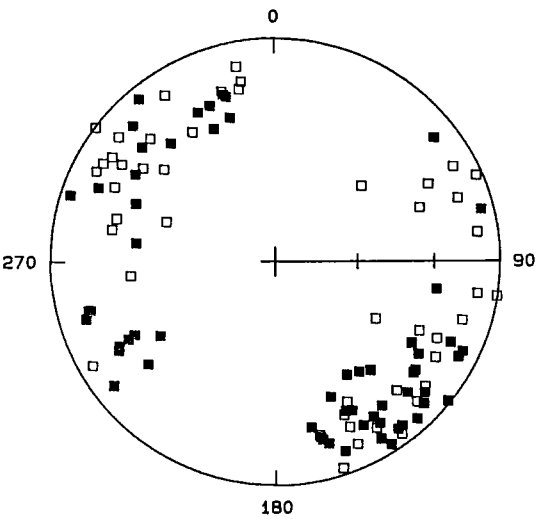


Fig. 4. Individual stable components deviating from A or A1, from the whole area investigated, are plotted together on an equal-area diagramm.

The rocks investigated were emplaced in the time range 337–309 Myear, i.e. in Late Visean to Westphalian times. As is shown below, acquisition of the magnetizations is mostly not related to crystallization of the rocks but to their later thermal and chemical histories. The rather good A and A1 clusters exclude significant relative rotations and tilting of the investigated sites after acquisition.

4.2. The A overprints

The A directions are found in all rock types, in the oldest Weinsberg granite as well as in the youngest aplitic dyke. Comparing the mean A VGP from the Southern Bohemian Massif (SBM) with results from other Paleozoic massifs of 'stable' Europe shows a clear consistency with the Late Permian (Saxonian-Thuringian) poles of southern France (Kruseman, 1962; Merabet and Daly, 1986; Merabet and Guillaume, 1988; Diego-Orozco and Henry, 1992) (Fig. 5). Very close to our result is the mean pole computed for the Bavarian Forest quartzporphyries from results from the Western Bohemian Massif (WBM) (Soffel and Harzer, 1991). The directions from the Bavarian Forest are probably Late Permian overprints, while those from the Fichtelgebirge are considered as Early Autunian primary magnetizations. The A pole is also in good agreement with results interpreted as Permian overprints in Early Carboniferous intrusives and volcanics from Vosges, Odenwald, Spessart (Edel and Wickert, 1991; Edel and Lewandowski, 1994) and in Early Paleozoic volcanics from Frankenwald (western Bohemian Massif; Kim and Soffel, 1982).

The same conclusion can be drawn for the A

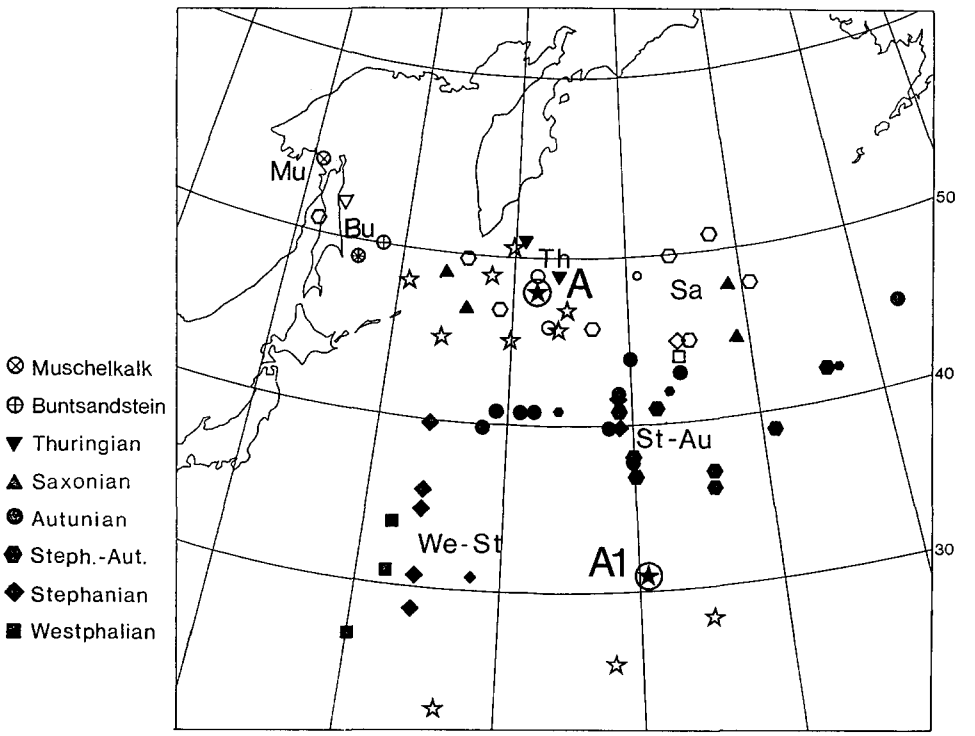


Fig. 5. Selection of VGP published for central Europe (France, Germany, the Czech Republic and Poland) for the Late Carboniferous to Early Triassic time span. Open stars correspond to overprinting of Devonian–Dinantian volcanics and intrusives from Vosges, Odenwald, Spessart (Edel and Wickert, 1991; Edel and Lawandowski, 1994) and Frankenwald (Kim and Soffel, 1982). Full stars indicate poles from this study.

magnetizations from the SBM. The originality of these overprints, however, depends on the type of remanence carrier. In the western massifs (Vosges, Odenwald, Spessart), the overprints are mostly carried by hematite resulting from low-temperature weathering processes occurring close to the surface (Edel and Wickert, 1991). Those from the Frankenwald also result from low-temperature oxidation but the carriers are (titano)maghemite (Kim and Soffel, 1982). In the SBM the carriers of the A overprints are predominantly sulfides, probably pyrrhotite. Under oxidizing conditions, pyrrhotite is easily altered to magnetite. Such conditions prevailed in Europe during the Late Permian as testified by the production of hematite in the western massifs and of maghemite in the Frankenwald. As this was not the case in the SBM, this means that the chemical process responsible for the A overprinting occurred at greater depth than in the previous cases. The area investigated was not brought to the surface in Late Variscan times but later, probably during the Alpine Orogeny. According to Frasl and Finger (1991), the basement outcropping south of the Pfahl fault was deeper rooted than the Bohemian part. Nevertheless, the depth has not exceeded 10 km, given the closure temperature of biotite of 300°C. The radiometric ages indicate that the temperature remained below that value after the Late Carboniferous.

4.3. The A1 overprints

The A1 directions also occur in different rock types, from the oldest 1B (BM 19, 21) to the youngest 2B group (BM 22, 26, 27). They are well developed in the quartzmonzodiorites south of the Pfahl fault (BM 22, BM 26 and its dykes), which belong to the latest intrusives. The carrier is mainly magnetite, partly sulfides. The mean A1 pole is intermediate to the published Westphalian to Autunian poles of Western Europe and Late Carboniferous overprints of Early Carboniferous volcanics and intrusives from Vosges, Odenwald and Spessart (Edel and Wickert, 1991; Edel and Lewandowski, 1994) (Fig. 5). The closest dated poles come from intrusives of the North Central Massif dated at 300 Myear and from a rhyolite of the central Black Forest dated at 296 Myear on the one hand (Edel, 1987; Lippolt et al., 1983), from Stephanian sediments of the central and eastern Bohemian Massif (Krs, 1968) and Autunian volcanics and quartzporphyries of the Thuringian Forest (Mauritsch and Rother, 1983) and the Fichtelgebirge (Erbendorf; after Soffel and Harzer, 1991) on the other hand. The larger scatter of these poles relative to the A poles is due to the fact that in Late Carboniferous–Early Permian times wrench faulting was still active and slight rotations may have occurred between and within the different massifs.

Previous investigations in different massifs of the Variscides have shown that the Early–Middle Carboniferous paleomagnetic directions were strongly rotated during the Late Visean–Namurian convergence, partly in relation with wrench faulting (Edel, 1987; Edel and Wickert, 1991). It is probably also the case for the SBM since the Pfahl fault was also a large dextral wrench fault in Devonian–Early Carboniferous times (Edel and Weber, 1994) so that the A1 magnetizations can be considered as secondary in the 1B and 2A groups. Concerning the A1 magnetizations from the 2B group, their origin is open to question. Assuming that the oldest ages of 309 Myear and 302 Myear obtained for the Freistadt and Schlägl granites, respectively (Friedl et al., 1992b; Frank and Schabert, unpublished work) are intrusion or cooling ages, the magnetizations may be primary. But these dates may also represent late events as is the case for the U/Pb dates on the Weinsberg granite (314–318 Myear, Von Quadt and Finger, 1991) and for the K/Ar ages from 2A and 2B granites (279–298 Myear; Frank and Schabert, unpublished work) which coincide with the Ar/Ar ages on muscovite from Danube and Rodl fault mylonites (281–294 Myear; Wallbrecher et al., 1991). Ductile deformation and recrystallization along the main faults means that in the vicinity of the faults (BM 22 and BM 26 for instance), conditions were favorable for magnetic overprinting. Hence, the A1 magnetizations of the 2B group can also be considered as overprints. Anyway, given the radiometric data and the correlations with other paleomagnetic data, a Stephanian age in the range

310–290 Myear is suggested for the A1 magnetizations.

4.4. The remaining magnetizations

The stable directions which diverge from A or A1 are scarce in the individual sites, hence no mean was calculated. The normal and reversed C clusters, obtained when plotting all directions from different sites together, however, raise some questions. The C magnetizations are often carried by magnetite (Fig. 4). Similar C directions were obtained from Late Devonian–Early Carboniferous rocks from the Vosges, Odenwald, Spessart (Edel and Wickert, 1991; Edel and Lewandowski, 1994) and the Harz Mountains (Bachtadse et al., 1983). It is tempting to consider these components as relicts of 'primary' magnetizations which have survived the A1 and A overprinting phases. If this were the case the SBM has undergone a large rotation in the middle Carboniferous. But, as some of the sites concerned show a northwest–southeast-striking foliation and/or lineation of the anisotropy of the magnetic susceptibility (AMS), consistent with the main Danube and Pfahl faults (others agree with

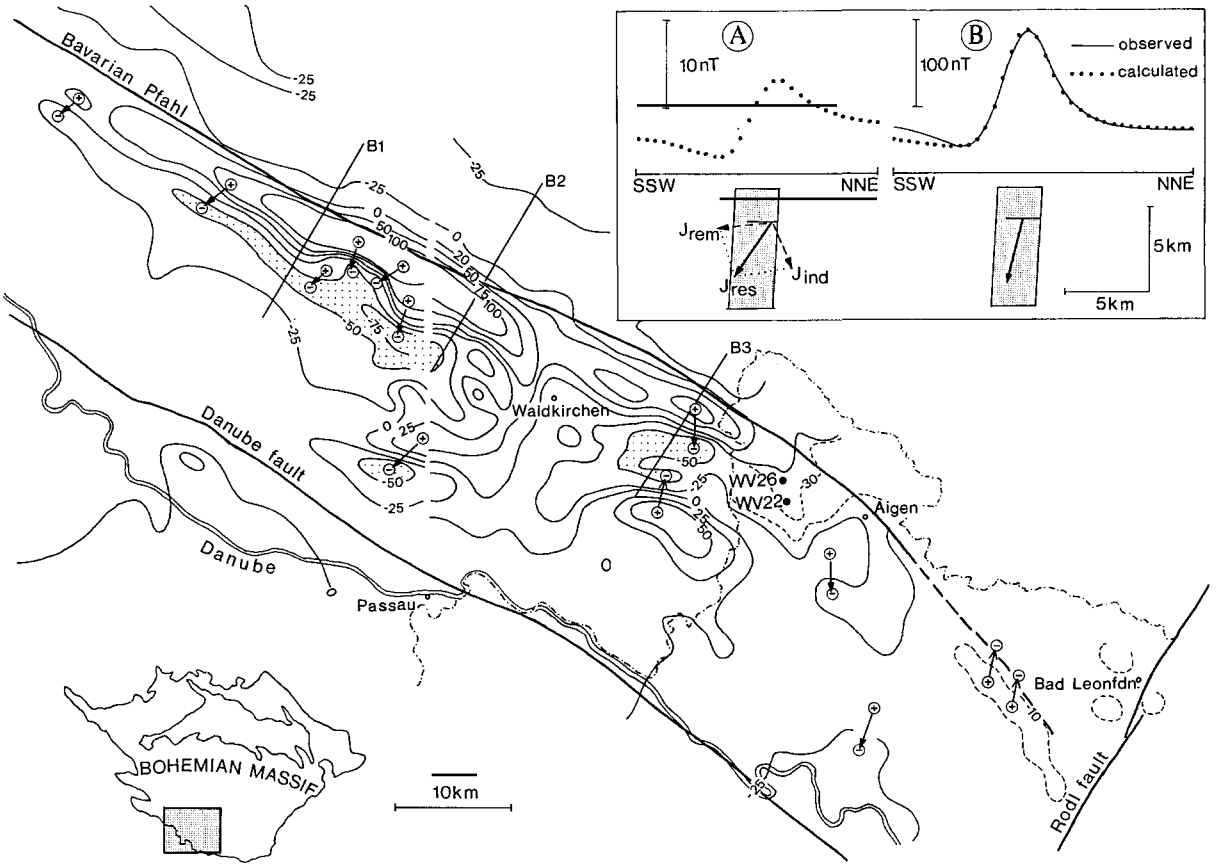


Fig. 6. Extracts of the aeromagnetic maps of Germany and Austria. The negative part (dotted) of the main anomaly is to the south, which indicates a significant remanent component of the magnetization responsible for the anomaly. Small (+) and (−) correspond to relative maxima and minima and the arrow gives the strike of the magnetization. (a) Magnetic anomaly computed with the same structure as (b) and the mean magnetization in the site BM 26; (b) modelling of a mean profile of B1, B2, B3.

north-northeast south-southwest strike of the Rodl fault), reflecting a remarkable deformation and influence on the magnetization directions the reliability of these magnetizations is not clear. The question of magnetizations older than A and A1 remains open.

5. Conclusions

Uniform directional patterns found even in outcrops with complex magmatic features indicate that chemical remanent magnetization (CRM) is the predominant type of remanence. This assumption is in agreement with petrographical studies revealing mineral changes in the temperature range of 200–400°C caused by hydrothermal processes (Jawecki et al., 1992) in granitoids of the South Bohemian Pluton.

The paleomagnetic sites being spread over a vast terrain which is cut into several segments by prominent shear zones (Fig. 1), the good grouping of site-mean directions corresponding with results from other parts of the Bohemian Massif leads to the conclusion that major block rotations did not concern the study area in post-Variscan time.

Overprinting lasted from the Late Carboniferous to the Late Permian, with two major phases, one in the Stephanian, the second in the Saxonian–Thuringian. The first phase can be correlated with several radiometric dates in the range 309–279 Myear and occurred in an environment of ductile deformation along dextral northwest–southeast and sinistral north-northeast–south-southwest strike–slip faults. The late remagnetization phase which is caused by hydrothermal processes, is contemporaneous with overprinting in other parts of the Bohemian Massif (Bavarian Forest, Thuringian Forest) as well as in other Paleozoic massifs of the Variscan belt (Vosges, Odenwald, Spessart).

The extension and amplitude of Late Variscan remagnetizations associated with recrystallizations in the SBM are illustrated by the magnetic maps of Germany and Austria (Fig. 6). South of the Pfahl fault runs a prominent, 73 km long magnetic anomaly which stands out with a minimum to the south which is due to a strong southerly remanent component of the magnetization. Outcropping rocks are granites and very small dioritic bodies. Two-dimensional modeling of a mean profile (corresponding to B1, B2, B3) leads to a 2.6 km thick, 0.7 km deep dyke, nearly vertical, and with a steep magnetization striking south–southwest (Fig. 6b). When taking the parameters of the mean induced and remanent magnetizations measured in the quartzmonzodiorites (BM 22, BM 26) which outcrop at the western extremity of the anomaly (Fig. 6a), and the same model as in (Fig. 6b), calculation produces a quite similar anomaly, but with a weaker amplitude. This shows that a nearly flat southerly remanent component corresponding to A, A1 or A + A1 overprints may explain the deviation of the magnetization from the present field direction. Consequently, the magnetic anomaly probably is due to Late Variscan recrystallizations within the magmatic basement. Fluids necessary for these mineralizations were carried by the close Pfahl fault.

In short, the granitoids of the SBP were emplaced in Early to Middle or Late Carboniferous times. The period of acquisition of magnetizations was significantly shifted and lasted from the Late Carboniferous to the Late Permian. The two main overprinting phases are rich in information on the tectonic and hydrothermal environment of the South Bohemian Massif, and on the whole Variscan belt. Further development of these data would require a close collaboration with petrographers and geochemists.

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